Electric Vehicle Market Performance & Insights

Research and Technical Methodology

Repository: https://github.com/manasgite54-cmd/EV-project

Purpose and Context

The project uses applied descriptive analytics. It transforms a source dataset into a cleaned data product, stores data in a structured form, explores distributions and relationships, visualizes selected measures and presents them through a Flask interface.

Detailed Project-Specific Content

- 1. Data provenance: identify publisher, license, access date, time/geographic coverage and record grain.
- 2. Preparation: use Python/Pandas for schema inspection, normalization, type conversion, missingness and duplicate checks.
- 3. Quality assurance: produce the JSON report, reconcile row counts and retain limitations for uncertain fields.
- 4. Database: use SQLite for local relational storage and SQL for repeatable queries; document actual table names, keys, types and import method.
- 5. EDA: calculate distributions, counts, medians/means where appropriate, group comparisons and relationships across available fields.
- 6. Visualization: build Tableau worksheets, a dashboard and a story with clear units, tooltips, source notes and caveats.
- 7. Web integration: use Flask routes and HTML/CSS templates for Overview, Dashboard, Story and Methodology.
- 8. Testing/reporting: validate outputs and links, record observed test status, discuss limitations and document reproduction steps.
- Interpretation: descriptive comparisons do not establish causality. Correlations among capacity, range, price and ratings are associations in observed records.
- Missingness can be systematic; a chart with complete cases may describe a different subset than the full dataset.
- Reproducibility requires repository code, SQL scripts, Tableau assets and setup instructions to match the report.

Project References and Artifact Locations

Project: Electric Vehicle Market Performance & Insights Repository: https://github.com/manasgite54-cmd/EV-project Processed dataset: data/processed/ev_market_clean.csv Reported processed records: 299,999 Cleaning report: data/processed/cleaning_report.json Tableau Dashboard: https://public.tableau.com/views/Book1_17931030431560/EVMarketOverview?:showVizHome=no&:embed=yes Tableau Story: https://public.tableau.com/views/Book1_17931030431560/Story1?:showVizHome=no&:embed=yes Accuracy note: the record count and artifact paths above are reported by the project owner. Exact schema, source provenance, units, calculated statistics, test results, and deployment status must be checked against the repository and actual artifacts. No unsupported results, approvals, or signatures are claimed.

Status: Detailed project-specific document. Any [TO BE PROVIDED], [TO BE VERIFIED] or [TO BE CALCULATED] item requires confirmation. No supervisor approval, signature, unverified test pass or successful deployment is implied.